

Deterministic Resolution and Continual Knowledge Revision from Raw TGI Experience

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Abstract

TGI needs a decision mechanism that turns source-valid geometric experience into a justified action or answer without interpolating an unsupported path. This manuscript specifies resolution over actual source-derived laws, complete joint state, context guards, and a verified frontier of available actions. Literal cases and whole-before word operators are distinguished; the latter are admitted only when every original organization binds the axis and every literal source replays exactly. A finite proof dictionary over source guards and goals supports exact congruence for the declared guarded-word system. Counterevidence interrupts a formerly supported route, while new actual experience can renew context laws without erasing the old event history. Complete owned transactions commit atomically and recover their certificates in a fresh process. V116 signed actual transitions, V119 source-owned local acquisition, and source-extracted regressions provide different evidence classes for this chain. The paper treats *NO_PATH*, *REVOKED*, and incomplete search as distinct outcomes; a search budget must not silently authorize a partial answer.

1. Research question and dependency

The earlier TGI mechanisms provide character identities, source-valid organizations, and interventional relations. None alone determines whether a new query has a supported route. Deterministic resolution asks: given the complete current experience, which result is entailed by every applicable source-derived law, what action should acquire a missing witness, and how does the decision change after actual counterevidence? TGI's founding goal [1] forbids replacing a missing route with a softmax guess. It also forbids counting a successful replay of an already stored sentence as general reasoning about an unseen query.

This paper owns the complete decision chain: measured context laws -> guarded word operators or literal cases -> finite joint frontier -> selected actual action -> new raw observation -> recomputed knowledge -> atomic checkpoint. It does not claim that the word-system proof covers every possible physical domain. Its significance is that each link is explicit, checkable, and revisable from original experience.

2. Source-derived operators and context guards

An actual transition records a complete before frame, raw action, after frame, source ID, port set, and ordering. Minimal context families are computed from original measured relations. A guard is a conjunction over the ports in one family. A transition law may be an exact literal case for one observed before frame or a whole-before operator such as prepend, append, interleave, or repetition. The operator is not inferred from the surface similarity of two examples alone. Its admission requires:

1. a phase-valid single-axis whole-before organization;
2. every axis bound and every original organization agreeing with the proposed operator;
3. exact action applicability and replay of every original literal source;
4. agreement of all applicable minimal context families on the **full concrete output**, not merely on an abstract region name.

If an operator, guard, or literal case is missing, the decision records a specific acquisition obligation. It does not create an implicit wildcard. The V116 guarded-word decision [2] details the admission conditions. This differs from memorizing a list of source-output pairs: the admitted operator can act on a new whole before string under a proved guard, while an unproved operator remains unavailable even if it would happen to predict a test answer correctly.

3. Finite proof dictionary and complete frontier

Let D contain all nonempty substrings of original source guards and declared goals. The V116 proof treats an input outside D as nonmember under an operator that preserves that whole input; a constant operator can instead return to a known exact word. This gives an exact finite congruence for the source-proved guarded-word system, not an unrestricted theorem about all future language. The state for planning is the complete joint collection of port regions and concrete values. All available actions, reachable regions, self-loops, return edges, and shortest alternatives are enumerated and checked independently [2,3].

A frontier certificate owns the search result. It may identify a complete supported goal path, a shortest acquisition request, multiple unresolved alternatives, or a closed no-path result within the declared state/action system. A ceiling that stops enumeration yields incomplete status, not a synthetic theorem of impossibility. Context-changing actions update the entire joint state before the next step is selected. This is important when two ports are correlated in the current experience but an actual separating action later shows they require distinct context families.

For an answer y , the constructive claim is conditional: $Certified(q, S, y)$ requires that every selected edge have a source-valid operator or literal witness, all competing applicable families agree on y , the complete route is within the verified frontier, and the current source history contains no revoking counterexample. If any premise fails, the system must preserve the obligation or refuse the answer. This certificate is the content of deterministic resolution; merely choosing the largest bond score would not satisfy it.

4. Actual experience and knowledge renewal

Selection initiates a real action through the external interface. The returned before/after frames are admitted as original NMU events, not as an expected-output fixture. Unexpected effects are retained as knowledge or opposition; they can interrupt the current plan and trigger a new frontier. Repeated support does not delete the counterexample. Later contexts may separate apparently conflicting laws, but that separation itself needs an actual distinguishing witness [4].

The owned continuation cycle [5] performs verified prefix, actual continuation, learning, complete-history renewal, and atomic commit. The checkpoint contains the original event history and the proof needed for a fresh-process recovery. A malformed input, stale external digest, or certificate that disagrees with the raw events is rejected before partial mutation becomes visible. This avoids a common form of false continual learning: a process whose in-memory state seems updated but cannot reproduce the same decision after restart.

The V119 local acquisition integration [6] adds source-owned updates above the prior native runtime and keeps evaluator-label substitutions outside the native input. The older integration decision is not used to assert every later capability by inheritance. The current source archive [7] and its extracted phase are the reproducibility anchor for the latest packaged runtime; older evidence retains its own frozen binding.

5. Falsification and result classes

The experiments separate logical completeness, actual feedback, and systems transport. In V116, a forged universal-input certificate is rejected; a past-only correlated context cannot erase the later separating experience; and unsupported guards remain acquisition obligations. Workspace and source-extracted audits each execute nine signed actual transitions across context exclusion, two-step context-changing goals, prepend, literal interleaving, distinguishing acquisition, counterevidence, and a newly acquired command [2]. The decision records 18 transitions and cold recoveries across both audits, 10 observed goals, two quantified exclusions, four retained replans, two counterevidence interruptions, and two unique context discoveries. The 34 captures preserve native input under evaluator-label substitution. These are counts of a declared panel, not an extrapolated frequency in arbitrary worlds.

The V106 continuation API [5] reports seven focused ownership/failure tests, 649 extracted regressions, 159 installed tests, and six physical transactions through actual cycles. V119 [6] reports 40 focused tests, 957 source regressions, 467 installed tests, four actual transitions and commits, four cold recoveries, and eight label-invariant captures for its integration scope. The present source distribution [7] runs its own complete phase and distribution checks from extracted bytes. These records should be read in their version order: later modules do not rewrite historical results.

Negative cases are essential. A no-path result after an unsupported guard is not counted as successful generalization. A changed action outcome is not silently normalized into the predicted state. A partial frontier produced by a resource ceiling is not rebranded as complete resolution. The paper's decision categories preserve these distinctions so that further learning can target a concrete missing witness.

6. Reproducibility and relation to NINMENI

NINMENI contributes the immutable NMU event substrate [8] and a research discipline that separates organization, memory, material update, and output capability [9,10]. TGI's guarded-law, frontier, actual-feedback, and atomic knowledge operators are its own derivations. The relevant modules are *tgi/guarded_word_closure.py*, its independent checker, *tgi/guarded_word_goal_frontier.py*, *tgi/guarded_word_goal_workflow.py*, and *tgi/local_acquisition_runtime.py*. Contracts and receipts [2–7] identify the source scopes and exact commands. The publication bundle contains the manuscript, source archive, frozen controls, and SHA-256 index.

7. Resolution procedure and certificate decomposition

At a fixed current source state, the procedure first verifies the complete original event ledger and phase validity. It then derives every minimal context family and literal law from those events. Candidate whole-before operators are admitted only after the four source tests in Section 2. The proof dictionary is built from the finite guards and declared goals, and all reachable joint-state edges are enumerated for the declared available actions. The frontier checker recomputes the same graph from source laws, compares all shortest alternatives, and checks each proposed goal path's original NMU lineage. The selector may then choose one certified action or an acquisition obligation. After an actual executor result returns, the complete process repeats on the enlarged ledger.

This order avoids a subtle circular proof. The expected answer must not be fed into the organization that later “proves” the answer. Nor may a selected action's anticipated effect be substituted for the actual before/after frames. The source-derived law and measured effect have distinct provenance fields. When they disagree, the disagreement is counterevidence and changes the knowledge decision. A fresh process recovery verifies the same sequence without consulting an in-memory shortcut or a hidden evaluator label.

Certificate layer	Owned premise	Typical falsifier
source/phase	raw event and valid organization	damaged or stale source receipt
operator	all bound axes and exact literal replay	forged universal-input witness
context	every applicable minimal family agrees	newly separating actual action

Certificate layer	Owned premise	Typical falsifier
frontier	all reachable declared edges and shortest alternatives	omitted self-loop or return edge
actuation	actual action/result chronology	expected outcome substituted for observed outcome
checkpoint	atomic ledger and certificate	wrong external digest or partial write

The statuses have operational consequences. *CERTIFIED* permits the declared route's output or next action. *NO_PATH* means no supported route in the completed declared graph. *REVOKED* means an earlier route's premise has been contradicted or its source has lost validity. *INCOMPLETE* means search, capture, or verification stopped before a decision could be proved. Combining these into a single “low confidence” number would lose information needed for the next experiment.

8. Continual-learning protocol and failure preservation

Continual learning is a chain of transactions, not a single successful append. The V106 cycle API [5] checks owned input/output, invalid plan, valid refusal, atomic failure, external digest, scope upgrade, and exact work boundaries. Six physical transactions include no-reset rollovers and an unexpected outcome. V116 [2] then tests actual separating experience, counterevidence interruption, retained replans, and a newly learned command under signed capture. V119 [6] re-bases local acquisition on the current native source while preserving the previous runtime modules. These are complementary rather than interchangeable tests: API recovery does not prove a context law, and an operator proof does not prove an atomic checkpoint.

The negative ledger is part of the method. A past-only prefix cannot resolve two continuations after the same prefix without another witness; false certainty there is a falsifier, not a minor quality reduction. Similarly, a completed test with its input instrument later found invalid must be retracted as an instrument failure. A resource exhaustion during proof construction yields no partial certified answer. Preservation of these cases is necessary to tell whether a later mechanism actually closes an obligation or merely hides it behind a new name.

References

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- [2] *contracts/GUARDED_WORD_INTEGRATION_DECISION_V116.md*.
- [3] *contracts/GUARDED_WORD_CLOSURE_PROOF_V116.md*.
- [4] *contracts/GUARDED_WORD_WORKFLOW_API_V116.md*.
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